Hidden Music: How The Revelation Of Hidden Environmental Processes Can Be Used In Generative Composition

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Abstract

Sonifying environmental data is an exciting way to influence generative music and reveal hidden processes in a public space. My installation work “Environmental Orchestra” aims to use environmental data, temperature, humidity, and CO2 concentration as well as plant activity to influence a piece of generative music with the intention of making the audience think more about these often-overlooked processes. My methodology consists of both practical and theoretical research into assorted topics around sonification as well as technical exploration, working with DIY electronics to create the piece and achieve the desired outcome.

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1 Introduction

The development and theory of generative music and its relation to the field of sonification opens up new opportunities for creatives to reveal hidden environmental processes, in novel ways. Through my research and creative development, I intend to explore this concept of “sonifying an environment”. I believe that the use of sonic installations and generative music can be used to display these processes that often go unnoticed as well as how they interact with each other and how human interventions can change them. My installation “Environmental Orchestra”, housed within the tropical greenhouse at Leicester University Botanical Gardens, represents a culmination of technical research. Mostly based on practical development of prototypes as well as building from other artists’ work. I also explore the broader topic of sonification to inform my approach. I utilise my skills of musical programming, generative composition and working with electronics to create the installation and complete my research; this paper discusses how I have done this in terms of practice as well as the decisions that informed my working.

2, Context

In this section I will lay out some important context for my work. Investigating pieces that experiment with the main themes of my installation allows me to better understand how it fits into the wider environment as well as influencing and aiding my development of it. All of the themes can be related to sonification, many of the pieces I use as examples do sonification in one way or another. This has been one of my findings through this research, much of generative music or sound art can be said to be “sonifying” something, be it environmental data or simply the output of a generative algorithm, this is perhaps due to the broad definition of sonification, a point I will explore later in this chapter.

The Site

A major consideration for my own work is the space it occupies, my decision to house the installation within the botanical gardens was partly circumstantial; the work required a space with plants in a somewhat enclosed environment. For this reason I would describe the piece as “site-type specific” meaning a work of art that requires a specific environment (in this case a greenhouse or perhaps garden) but not a specific locale, I had no intention to comment on Leicester botanical gardens specifically, only the environment within a greenhouse.

A site-specific work is described as “inextricably linked to the locale”(Rosenthal, M. 2003 p.28) Rosenthal goes on to clarify that for a site-specific work “Moving it is impossible, since the work cannot be understood or seen except in relation to the place.”(2003 p.28) an excellent example of a site-specific work that would fall into this definition would be Robert Smithsons’ 1970 work “Spiral Jetty”, located in the Great Salt Lake in Utah. it is described as “A place where the viewer can re-experience nature’s vast scale” (Renshaw, 2013, p.55) this piece both relies on its environment and interacts with it, the water level of the lake changes how the work is perceived, during times of high rainfall the spiral can be completely flooded and during drought it becomes clear again.

Another interesting take on the site as artwork is Dan Grahams’ “March 31, 1966”, this is perhaps best described as a self-referential poem. Describing a specific place but it provides an interesting example of a work of art that is intrinsically linked to a certain space. An example of a site-specific piece that also interacts with the environment sonically would be Mike Tonkin and Anna Lius’ “Singing Ringing Tree.” This is a a site-specific aeolian sculpture. This piece demonstrates the use of sonification, which I will talk in-depth about this in a later chapter but this piece is an interesting example as rather than using data gathered from sensors and then processing to sonify it (As my piece and many others do), this sculpture uses the physical process of wind blowing through pipes to sonify the environment.

Considering the space was important throughout the development of my piece, both when it comes to how the work influences the space but also when it comes to the technical restraints of the space, having the context of previous works has helped me understand how my installation fits into its environment.

Revealing the Hidden.

Revealing hidden processes is one of the core aspects of my piece. By exploring how others have gone about this as well as their theory ,I can better understand what it means to reveal the hidden through sonification and installation.

Bill Fontanas “Sonic Shadows” explores this concept. The piece (constructed at the San Francisco Museum of Modern Art) “explores both visible and invisible features of the museum building.” (SFMOMA, 2020) Through “revealing” various sounds that the museum building itself produces. Fontana achieved this by attaching accelerometers and vibration sensors to various sections of piping in the boiler room as well as a pedestrian bridge within the museum and amplifying their signals. This could be seen as a form of sonification, the “sounds” already exist in the form of vibrations in metal etc. but many of them are inaudible. By revealing these sounds, as well as taking them away from their specific origin (the boiler-room is located in the basement while the sounds are played in the space above) Fontana re-contextualises them into a piece of music “created by the building”.

Another piece which arguably experiments with the idea of revealing the hidden processes around us is Janek Schaefers’ 1995 piece, “Special Delivery”. Here a cassette recorder was posted across London and recorded the loudest sounds along the way, resulting in the final piece of selective field recording. while conceptually very different and perhaps not considered sonification this work allows the listener access to the journey, while writing about the piece Schaefer notes “it was an exciting experience discovering the sounds for the first time”(2001) This idea of revealing the pieces journey through sound follows my own interests, although this piece reveals a hidden “man-made” process (the postal system) while I reveal hidden natural processes.

These pieces play with my interest of taking overlooked occurrences and bringing them to the forefront. While each piece has a different reason for revealing its process and all do so in different ways, they all use it as a core concept.

Environmental Integration

An important factor of “Environmental Orchestra” is, of course, the environment. While for this piece I’m not really considering factors such as climate change but simply using the environment to generate music, it’s still important to consider pieces that use ecological data to demonstrate these issues.

Integrating the environment into artworks can serve a range of purposes, as mentioned the environment can inform or even be the centre of the piece such as in “Spiral Jetty” but the piece can also be used to create a narrative such as in Rachel Jacobs’ environmental work witch I explore in a later section. Another particularly relevant example is “Dendrophone” by Peter Batchelor, described as a “sound installation that transforms invisible environmental data into an immersive listening experience” (Batchelor, 2025) this concept is very similar to

my own. “Dendrophone” also utilises similar environmental data, humidity and carbon dioxide levels though uses it in a somewhat contrasting way “The installation is designed to blend with the natural sounds of the forest, inviting listeners to experience these in a new way.”(Batchelor, 2025) this is fairly different to my piece as I intentionally used synthetic sounds to generate music rather than specifically reflect the sounds of the environment.

The use of environmental data has also been used to draw attention to the issue of climate change, the piece “A Song of Our Warming Planet” by cellist Daniel Crawford

> “transformed 133 years of annual global temperature anomalies into a haunting, atonal melody that stretched across almost all of the instrument's range.”(George, Crawford and T. Reubold, 2014)

This is a fairly simple way to draw attention to the issue of climate change through music, simply mapping the temperature data to pitches creates an effective and powerful representation of the trends.

Overall my research into how the environment can be used to drive music has allowed me to both understand where my piece fits but has also informed many of my decisions during its development, while I am mostly using the environment around the installation to drive music I would like to use my system to explore issues such as climate change in future versions.

Sonification

Perhaps the main theme of my piece is sonification. Interpreting environmental data in order to generate music. As a result, I have conducted extensive research into sonification, both its theory and several specific pieces that I think demonstrate the process with a variety of intentions.

Sonification is described as “synthetic nonverbal sounds [that] can represent numerical data and provide support for information processing activities of many different kinds.” (Barrass and Kramer, 1999). While helpful and widely accepted, this definition remains open to interpretation; as Lindborg Puts it in his 2023 paper “sonification is still an emerging discipline that struggles to define its boundaries”(p. 02), For example if I were to represent temperature by using a sine wave whose pitch correlated directly to the current temperature in degrees C this would be classed as sonification. This method is what I would describe as an “audible graph”, a clear and direct representation of data with very few creative constraints; if the pitch is higher that means the temperature is higher etc. This method may be helpful in interpreting data: a similar method to this has been used to aid people with visual impairments understand graph data(Dobosz, 2024) .And the graphing program Desmos also implements a function that allows the user to sonify a graph (Desmos, 2024). These methods generally lead to uninteresting musical results. A different technique would be more similar to my experiments during the start of the year and my “In C” performance, using spikes in “plant data” almost as a “trigger” for randomness. Perhaps this type of sonification could be achieved by a system that measures the temperature of a space and if the temperature has increased by a certain amount, commences a new pre-composed musical phrase. This method has the advantage of always sounding aesthetically interesting; however, the relation between sound and data is far more obscure, I would call this “data-

influenced composition". Of course, most works straddle this line between data-influenced composition and sonic graph, a good example of this be Ian Baxters' "Playing the Weather". Baxter describes the process of the piece as

> "pitches [are] derived from a reading a newspaper weather report, assigning pitches to the names of the cities in the 'around the world' weather table, and moving through the report based on the reported temperatures. Using a chance element of this kind means there can be no pre-determination of what the piece would sound like on any given day."(2020)

While in many ways this is similar to the audible graph, Baxter *has* decided on the sound material, mappings and scale so it is still at least somewhat influenced by the composer.

Finding a balance between these two extremes of sonification, that is entirely data-driven sound, with little consideration to aesthetics and "data-influenced composition", where aesthetics is one of the main considerations, is a major point of discussion within the field. As Paul Vickers argues

> "Music and sonification have ostensibly different goals. The composer strives for aesthetic interest [while] In sonification, it is not aesthetic interest but successful signification of the data that is the goal. But are these goals mutually exclusive and, if not, how can the practice in one field inform the other?" (Cobussen, Meelberg and Truax, 2016, pp.135–145)

This perhaps summarises one of my main goals with the piece, is it possible to create an installation that is both aesthetically interesting and successfully represents the data?

This issue of data / sound relationship could also be said to the perennial issue in generative and process art in general. While the process may technically decide the end piece, the artist ultimately has control over the constraints. John Cages' "Variations III", for example, intends to remove the composer as much as is possible, allowing chance to create the piece. However, even here the composer has the final say on sound material, tempo, pitch etc, it is impossible to completely remove the composer or performer from the composition. What is important for my piece is that I choose a position on this scale between audible graph and data-influenced composition, I'm not interested in representing the data perfectly but want to allow the listener to understand the processes somewhat intuitively, or "make the scientific data "come alive"".(Lindborg 2023) This means I will implement generative techniques to interpret the data and use my own judgement to convey the "feel" of the data; for example if the temperature is rapidly changing perhaps the sound should be chaotic as this could be read intuitively by the listener. This idea is echoed by Jacobs when she mentions that climate data is "abundant yet inaccessible and misaligned with the needs of different audiences." (2017) and that it would be more helpful to communicate this data through more " sensory, tangible and visceral representations" (Lindborg 2023) Jacobs, Howarth and Coulton explore how this has been done successfully in their paper, discussing artworks such as Jacobs' 'A Conversation between Trees'. Here various works engage the public with "playful, visual and sensory interactive experiences" (Jacobs et al., 2013) in order to better communicate climate data.

Generative Music

Generative Music is another major theme of my piece. I already have experience in creating generative compositions and so much of my research was done as part of the build process. That being said context around the practice is still important to understand and sets the scene for my piece.

Generative music, and generative art is described as “works that are ‘generated, at least in part, by some process that is not under the artist’s direct control” (Collins, 2008) this is a similar concept to that of process music, “Pieces of music that are literally processes” (Reich and Hillier, 2002, pp.34–36) Steve Reichs’ “Pendulum Music” is an example of process music, in this piece microphones are suspended above loud speakers and swung, generating feedback as they pass over the speakers. This is a good example of how music can be created through process. The term “generative music” is generally attributed to Brian Eno and his work. Enos’ “Music for Airports” as well as his work with the SSEYO Koan system pioneered generative composition. Eno was also one of the first to experiment with computer generated music, when talking about his process Eno notes “Generative music […] specifies a set of rules and then lets them make the thing.” (Eno, 1996) while artists like Reich worked with physical processes Eno is known for creating these rule sets or “systems” using software like that developed by SSEYO to generate music. This is more similar to my approach to generative composition, for “Environmental Orchestra” I use Puredata patches to program my systems for music creation.

Overall, the context I have gathered has provided me with valuable theory and background, both grounding my work and providing me with inspiration. While the pieces that I have discussed here are often very different to my own installation they all have aided me with decision making or helped me to understand the various factors of my piece.

3, Theoretical Framework

This section lays out some of the theory behind the piece. Here I explore the presidents around bio-sonification as well as how the various systems work.

The purpose of this installation is to reveal “hidden Process” through sonification, that is to collect data from a space and use it to inform the music. I wanted to do this in a way that allows a listener to intuitively understand processes occurring within the space. Specifically, it focuses on environmental data, incorporating Biodata from plants as well as more general data from their vicinity (temperature, humidity and co2 concentration.)

Many of the biodata sonification examples I have found operate more on the side of data-influenced composition, Sam Cusmanos work for example constrains the played notes to a scale. This is something I have implemented in my piece as it is possibly the easiest way to turn random or seemingly random impulses into aesthetically passable music. If the simplest code for generative music is a random note generator, the next simplest is a random note generator quantised to a scale.

For this installation, a major consideration was how I could gather data effectively while keeping to a budget. This immediately meant I would be working with DIY electronics such as Arduino, Raspberry pi etc. I knew I wanted to collect some form of “Biodata”, that is data collected from plants. The most common way that I found to do this relies on the idea that there are “microcurrent fluctuations” within a plant’s leaves that can be used to determine activity. This theory can be traced back to the experiments of Cleve Backster, a polygraph expert, involving plants and polygraphs. Much of Backster’s work has been questioned (his ideas around plant consciousness and his theory that plants can seemingly read human minds and “detect the termination of life”(CyberFox, 2020) have been debunked by many (Galston and Slayman, 1979)). However, the idea of collecting some form of data from plants using his method of electrodes attached to leaves has since been used by artists for biodata sonification. Whether or not this is an accurate way to measure plant activity is, at least for this installation, somewhat irrelevant; I am intending to create a creative artistic piece, not a perfect scientific method of data collection. In many ways, the *idea* that the sound is created by the plants is more important than whether or not it actually is. I am using plant activity as a creative tool to influence generative music, not collecting this data for research into the plants themselves. . That being said, through my own experimentation there does seem to be some correlation, certain times of day seem to cause the plant to be “more active” and I have even been able to detect a change in activity when turning on/off a grow lamp though these results prove hard to reproduce.

The circuitry used for the installation is effectively a very simple “lie detector” circuit utilising a “555” Timer, this circuit produces pulse waves, the width of which is determined by the resistance across two electrodes. A change in pulse width indicates a change in resistance and so, in theory, if these electrodes are connected through a plant, any change in conductivity across the plant can be detected. This theory is laid out by Sam Cusmano and built on by others such as Kurundkar et. al, who claims that “plants in different conditions and habitat react and behave in different manner” (2023). It has also been used in the “MIDI

Sprout” device(MIDI sprout n.d.) I have my reservations about the effectiveness of this technique; however, my results do largely follow the findings of others. Again, I intend to use the plant data as an artistic tool, so discovering if these fluctuations mean plant activity or if they are simply random noise is outside the scope of this project and largely irrelevant; I am using other data which is far less subjective (temperature, humidity, co2 concentration), so the idea of “plant activity” is simply part of a larger picture.

4 prototypes and development

In this section I will explain my process developing “Environmental Orchestra” the various prototypes and how they influenced the final piece.

Biodata sonification is something I have been experimenting with for some time. During the first module of the year, I started work on the data collection circuit and code implementing this in a Max MSP patch. While this patch did utilise plant data it only used it as a simple trigger; that is, if the plant activity breached a certain threshold a note would be triggered. While this would technically count as sonification As discussed in section two, the relationship between sound and data is fairly loose, while the plant data does impact the music it only does so in a controlled and minor way. It was also at this time that I began to think about other data that I could use; this is when the idea of sonifying the environment around the plants began to take shape.

The next step I took involved multiple plants, I decided that I would create a sort of “plant quartet” and use the triggers from the plants to play a piece, specifically “In C” by Terry Riley. This experiment served several purposes, I wanted to test the validity of chaining several sensor circuits to a single Arduino as well as see how different plants react, my findings were that the Arduino had no issues with this setup and that different plants reacted fairly differently, for example my “birds nest” fern was fairly active with readings changing often and by larger amounts when compared to my Rubber plant. This gave me some hope that my circuits were working as if they were simply picking up on random noise; there would be little difference between the plants. The piece itself was admittedly not particularly interesting. I transcribed the piece to midi and used a max4live patch to change to a different clip every time the plant caused a trigger. I do think that using this method to play “In C” is in keeping with the intention of the piece, removing compositional control and giving it over to a natural and unpredictable process.

Once happy with my plant data experiments I went on to investigating other environmental data. Specifically, temperature, humidity and co2 concentration which I gathered from two sensors, the AHT20(temperature and humidity) and SGP30(co2). I decided to use the Adafruit break-out versions of both these sensors as they are relatively inexpensive and have robust documentation, allowing me to get them working quickly. Both these sensors communicate over the I2C protocol; this makes it very simple to set them up and start gathering data.

Once I had the environmental data collection worked out, I began planning ways that this data could be sonifyed, I was mostly focussing on learning to use this data within pd so was mainly testing the reliability of the comport external and patch I was using. My initial patch involved a receiving the data from the Arduino and testing various ways it could be used sonically. This is where decisions such as sound material started. I knew that I would have to run the synths on a raspberry pi and so wanted to keep them simple, this lead to the decision to create a sonic environment that would be somewhat similar to the album “Plantasia” by Mort Garson, organic sounding synths seem appropriate for the installation and are easy to recreate with limited synth design. One aspect of this first patch involved the idea of measuring how much a value had changed recently, for example I created a system that checked the current humidity against the humidity at the previous reading, if it was

higher a counter would increase if lower the counter would decrease and if it hadn't changed significantly the counter would tend towards zero. This gave me a single number that represented if the humidity was increasing or decreasing and how long it had been doing so for. I also used the same system to detect a change in plant activity. This initial patch was more of a proof of concept than a true representation of the final system but provided me with several useful ideas, one being the concept of representing the energy of a space, as it changes over time, rather than just raw data. For example, if humidity was high but stable this would be a different situation to if humidity had rapidly increased and so the sound energy should reflect this.

5, Realisation

In this section I explain how the final installation works as well as evaluate its effectiveness on the day.

After building the data collection systems the next stage was deciding how this data could be used. Methods like those used by Cusmano and the MIDI Sprout collect the data, convert it to MIDI notes, usually quantised to a scale, and then send this data to a synth. While I experimented with this it became apparent that, for the installation, it would likely be too cumbersome as the botanical gardens were wary of long cable runs and there was only one plug socket in the greenhouse. For this reason, I decided installation had to run on battery power. In order to create a (mostly) wireless system the software needed to run on a raspberry pi zero as they were my best choice for low power computers with appropriate, all be it limited capabilities. In addition to this consideration, converting the data to MIDI immediately restricts how it can be used, once the data has been converted into MIDI information it becomes harder to detect changes, trends and fluctuations. As well as this keeping the data “raw” allows for its use in non-MIDI controls such as filter cutoff. For these reasons I decided to use Puredata (pd) to interpret the data, as it allows me to gather the raw information from the Arduino and process it in any way, while remaining lightweight enough to run on the Pi.

Getting the information from the Arduino into a pd patch was accomplished using the “comport” external (Puredata.info, 2017) and a patch from a corresponding forum post (Forum.pd, 2020). This allows me to send several pieces of information, in a string format, from Arduino to pd and have pd interpret it correctly. This is important as it means that I can send all environmental data at the same time as one string.

The University of Leicester botanical gardens seemed to be a good place to present the installation, specifically the tropical greenhouse. This space appealed as its enclosed, relatively small and has several large plants. However, this space has a few downsides, the humidity was a concern as I worried about the various electronic components being damaged by water. As mentioned is also only one plug socket available and no internet connection, The installation comprises several speakers scattered around the greenhouse, but It is important that they don't have a wired connection to each other or to some sort of central hub such as a laptop for health and safety reasons, I can’t run long extensions for power, data or audio. For this reason, each “speaker” must act individually or in pairs to reduce cable runs. This posed a problem: how can the environmental data (temperature, humidity, co2) be shared wirelessly across multiple machines without an internet connection. I didn’t want to use multiple sensors as the cost would add up quickly so decided that I would utilise the ESP32 board. This is a microcontroller, like an Arduino, but has Bluetooth and Wi-Fi capabilities built in. It also has its own protocol for wireless peer-to-peer communication, ESPNOW (Espressif, n.d.). This uses the Wi-Fi protocol but does not require a connection to router or hub. Setting up a “one-to-many” network using ESPNOW allows me to gather environmental data using one microcontroller and then send that data to all the others as a string; the others can then add the plant data they gather to this string before sending it to a raspberry pi running my pd patch. While this may be a somewhat convoluted system it does work, As well as this it means that there is no central hub so the project would work in any

space and be scaled up by simply creating more “modules”. One issue with this method is the difference between Arduino and esp32 boards logic level voltage . Arduino are capable of outputting 5v while esp32 can only output at 3.3V. This isn’t an issue for the AHT20 or SGP30 boards as they can operate at either; however, the 555 timer requires 4.5V to operate. I had seemingly no real issues running the timers on the lower voltage but in the interest of reliability I switched them out for a lower power variation, the TLC555 which can operate as low as 2V. With audio I was able to simply use AUX cables connected to each Raspberry Pi, this was acceptable as the cables are small and not in any walkways. I did consider the use of a wireless audio system and may implement one in future, but all the methods were either reliant on line-of sight (IR receivers) Wi-Fi or were simply too expensive or didn’t have adequate reliability / range.

Installation day

After setting up the installation in the greenhouse there are several points I would like to consider. Overall, the sound and data aspects of the piece worked, nothing broke during the three hours that it ran for, and the music produced was aesthetically pleasant and, in my opinion, and that of people who saw the installation changed enough to remain interesting. The piece was not without its flaws however. During setup I wanted to space the modules out widely through the space, creating a more “immersive” sonic environment, when I attempted this one of the ESP32 boards wasn’t receiving the environmental data. While the ESPNOW protocol is advertised as having a range of up to 200 meters, and I had tested the range myself, I believe the boards being housed in plastic boxes, hidden under plants, significantly reduced the range. In future I intend to use either the boards “Long range mode” which allows for higher range or external antenna, both of these would likely be essential if I were to scale-up the installation for example in an outdoor scenario. Despite this the spread of the sound was still acceptable, I was sure to level all the speakers and kept them fairly quiet so the piece wouldn't be overwhelming. Throughout the duration of the installation there were several interesting things I noticed, there were points where I had the door open and so humidity and temperature both decreased, this made a noticeable difference in certain elements of the music. As well as this I noticed that standing near the sensor module led to a change, likely due to my breath increasing CO_2 in the area.

Overall, the installation was a success, at least in proving that my system works. I would like to set up the piece again but run it for longer in a more inconsistent environment, having the piece running outside over the course of a week for example would lead to more dramatic changes in the music and would be more pronounced. Due to how the system works it could be easily adapted to work outside, as I mentioned I would need to sort out the range issue but also likely use larger capacity batteries.

In total around forty people saw the installation, several asked what it was about and commented on it, mostly the interest was on the plant data collection and how the sound related to this. If I were to setup the piece again, I would have included more information about it on posters etc, I did have one on the entrance to the greenhouse, but few people saw it.

6. Conclusion

Overall “Environmental Orchestra” achieved my aim of sonifying an environment using DIY techniques and “revealing the hidden processes” that occur around us. As well as this it allowed me to research how public installations can work, organising with the botanical gardens, building a functioning piece to a deadline and conducting my own practical research will all aid me in future installation work. There are several things that could be developed and further researched however, the actual sonification element was the last element I considered and as a result the link between data and sound could be better thought out. However it still allows the listener to get an idea of how the environment around them is changing. Through researching topics around sonification and generative music I have encountered several questions that represent perennial issues in the fields, sonification is seen by many, myself included, as an excellent way to spread awareness of topics such as climate change, as Lindborg put it “Sonification,[...], and visualization, offer techniques for representing climate data with often innovative and exciting results.”(2023) however as I have discussed previously (and discovered myself during my own work) it is difficult to get a balance between data representation and sound. While I have worked extensively with generative music in retrospect, I wasn't able to fully explore how different generative techniques can be used in conjunction with real-world data, this is simply due to time limitations, having a working and reliable installation was more important than fully investigating this. This was a common problem with the installation progress, the technical side often took longer than the creative side, despite this, now I have a working system I intend on using “Environmental Orchestra” as a platform for future projects, the hardware and software that I have developed could easily be modified for a number of different projects or used as a base for future experiments.

Word Count: 5550

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Appendix

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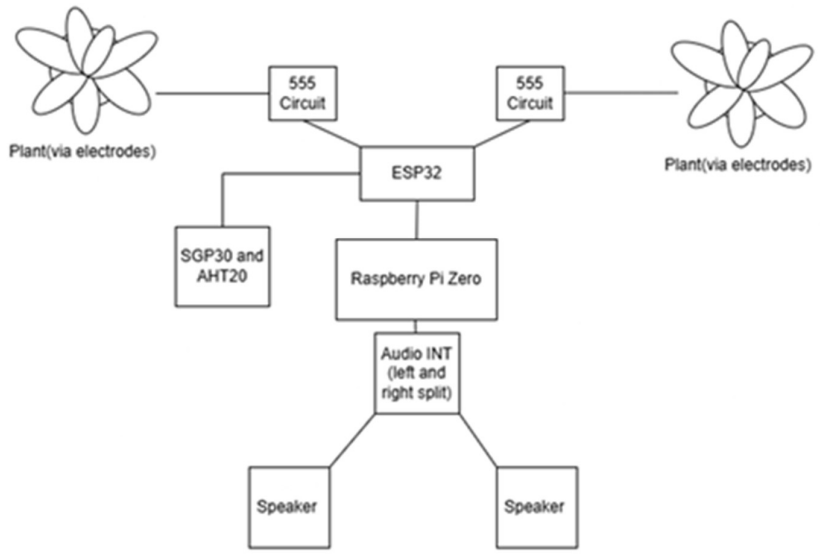

Figure 1 Diagram showing how each "module" is put together

Figure 2 Image of module one in situ during the installation

Figure 3Image of module two in situ during the installation

Figure 4Image of module three in situ during the installation